Chapter 1 — Key Terms Glossary, Self-Test, and Answer Key

Part 1 is a complete glossary of every key term with the textbook's definition (lightly compressed) plus a plain-English restatement. Part 2 contains the chapter's own practice questions ("Show What You Know" and "Test Prep — Are You Ready?") with the official answers from the book's Appendix C. Part 3 is a set of extra self-quiz prompts built for spoken review — perfect for a podcast to quiz the listener.

PART 1 — GLOSSARY (A to Z)

Behavioral perspective — An approach suggesting that behavior is primarily learned through associations, reinforcers, and observation. *Plain English: you act the way you've been trained by your environment.* (Watson & Skinner; strong nurture position.)

Behaviorism — The scientific study of observable behavior. *If you can't see it or measure it, don't study it.* (Founded by Watson.)

Biological perspective — An approach that uses knowledge about underlying physiology (hormones, genes, brain activity) to explain behavior and mental processes. Related field: **neuroscience**, the study of the brain and nervous system.

Biopsychosocial perspective — Explains behavior through the interaction of biological, psychological, and sociocultural factors. *The everything-at-once lens.*

Case study — A type of descriptive research that closely examines an individual or small group, often gathering information through many avenues (interviews, questionnaires, records). Invaluable for rare events; a sample of one, so it cannot test hypotheses or support generalizations.

Cognitive perspective — An approach examining the mental processes that direct behavior (thinking, memory, language). Offshoot: **cognitive neuroscience**, which searches for physiological explanations of mental processes, especially in the brain.

Confounding variable — A type of extraneous variable that changes in sync with the independent variable, making it difficult to discern which one is causing changes in the dependent variable. *The saboteur that moves in lockstep with your treatment* (e.g., time of day differing between caffeine and control groups).

Control group — The participants in an experiment who are NOT exposed to the treatment variable; the comparison group.

Correlation — An association or relationship between two (or more) variables. Positive = both move together; negative = inverse.

Correlation coefficient — The statistical measure (symbolized as r) indicating the strength and direction of the relationship between two variables; ranges from –1.00 to +1.00; closer to the extremes = stronger, closer to 0 = weaker. [Synonym: Pearson's correlation coefficient.]

Correlational method — A type of descriptive research examining relationships among variables; enables prediction but cannot determine cause and effect.

Critical thinking — The process of weighing various pieces of evidence, synthesizing them, and evaluating the contributions of each; disciplined thinking that is clear, rational, open-minded, and informed by evidence.

Debriefing — Sharing information with participants after their involvement in a study has ended, including the purpose of the research and any deception used.

Dependent variable (DV) — In the experimental method, the characteristic or response that is measured to determine the effect of the researcher's manipulation. [Synonym: response variable.] *The outcome you measure.*

Descriptive research — Research methods that describe and explore behaviors, but whose findings cannot definitively state cause-and-effect relationships. Types: naturalistic observation, case study, survey method (and the correlational method is also descriptive).

Double-blind study — A study in which neither the researchers administering the independent variable nor the participants know what type of treatment is being given. Controls both participant expectations and experimenter bias.

Evolutionary perspective — An approach that uses knowledge about evolutionary forces, such as natural selection, to understand behavior: thoughts, emotions, and behaviors are shaped because inherited traits increase in frequency when adaptive and decrease when maladaptive. (Based on Darwin's theory; David Buss is a modern founder.)

Experiment — A controlled procedure involving careful examination through scientific observation and/or manipulation of variables.

Experimental group — The members of an experiment exposed to the treatment variable or manipulation; the treatment group.

Experimental method — A type of research that manipulates a variable of interest (the independent variable) to uncover cause-and-effect relationships. The only method that can establish cause and effect.

Experimenter bias — Researchers' expectations that influence the outcome of a study. [Synonyms: experimenter effect, researcher expectancy effect.] Classic example: Clever Hans.

Extraneous variable — A characteristic of participants or the environment that could unexpectedly influence the outcome of a study.

Functionalism — An early school of psychology that focused on the function of thought processes, feelings, and behaviors — how they help us adapt to the environment. (William James; inspired by Darwin.)

Humanistic psychology — An approach suggesting that human nature is by and large positive, and the human direction is toward growth. (Rogers & Maslow.)

Hypothesis — A statement that can be used to test a prediction.

Independent variable (IV) — In the experimental method, the variable manipulated by the researcher to determine its effect on the dependent variable. [Synonym: explanatory variable.]

Informed consent — Acknowledgment from study participants that they understand what their participation will entail (procedures, risks, benefits, confidentiality) — the participant's way of saying "I understand my role in this study, and I am okay with it."

Institutional Review Board (IRB) — A committee that reviews research proposals to protect the rights and welfare of all participants. All experiments on humans and animals require IRB approval.

Introspection — The examination of one's own conscious activities. (Wundt's method; also used by Titchener's structuralism.)

Natural selection — The process through which inherited traits in a given population either increase in frequency because they are adaptive or decrease in frequency because they are maladaptive. (Darwin; basis of the evolutionary perspective.)

Naturalistic observation — A type of descriptive research that studies participants in their natural environments through systematic observation; the observer must not disturb participants or their environment.

Nature — The inherited biological factors that shape behaviors, personality, and other characteristics.

Nurture — The environmental factors that shape behaviors, personality, and other characteristics.

Observer bias — Errors in the recording of observations, the result of a researcher's value system, expectations, or attitudes. Reduced by using multiple observers.

Operational definition — The precise manner in which a variable of interest is defined and measured.

Placebo — An inert substance given to members of the control group; the fake treatment has no benefit but is administered as if it did. The *placebo effect*: expecting improvement can itself produce improvement.

Population — ALL members of an identified group about which a researcher is interested.

Positive psychology — An approach that focuses on the positive aspects of human beings, seeking to understand their strengths and uncover the roots of happiness, creativity, humor, and so on.

Psychoanalytic perspective — An approach developed by Freud suggesting that behavior and personality are shaped by unconscious conflicts.

Psychologists — Scientists who study behavior and mental processes.

Psychology — The scientific study of behavior and mental processes.

Random assignment — The process of appointing study participants to the experimental or control groups, ensuring that every person has an equal chance of being assigned to either. (Happens *within* a study — don't confuse with random sampling, which selects who is *in* the study.)

Random sample — A subset of the population chosen through a procedure that ensures all members of the population have an equal chance of being selected to participate.

Replicate — To repeat an experiment, generally with a new sample and/or other changes to the procedures, the goal of which is to provide further support for the findings of the first study.

Representative sample — A subgroup of the population selected so its members have characteristics similar to those of the population of interest; required for generalizing findings.

Sample — A subset of a population chosen for inclusion in an experiment.

Scientific method — The process scientists use to conduct research, which includes a continuing cycle of exploration, critical thinking, and systematic observation. Five steps: develop a question; develop a hypothesis; design study and collect data; analyze the data; publish the findings.

Sociocultural perspective — An approach examining how social interactions and culture influence behavior and mental processes. (Vygotsky.)

Structuralism — An early school of psychology that used introspection to determine the structure and most basic elements of the mind. (Titchener.)

Survey method — A type of descriptive research that uses questionnaires or interviews to gather data.

Theory — Synthesizes observations in order to explain phenomena and guide predictions to be tested through research; a well-established body of principles resting on a sturdy foundation of scientific evidence — NOT a guess or hunch.

Third variable — Some unaccounted-for characteristic of the participants or their environment that explains the changes in the two correlated variables. [Synonym: third factor.]

Variables — Measurable characteristics that can vary over time or across people.

Bonus (not on the key-terms list but used in the chapter): **empirical evidence** (data from systematic observation or experiments), **descriptive statistics** (organize and present data), **inferential statistics** (make inferences and determine probabilities beyond the data set), **empiricism** (Aristotle: knowledge through sensory experience), **dualism** (Descartes: body and mind as two separate interacting entities), **social desirability** (bias toward answering in ways that look good), **peer review** (expert scrutiny before publication), **neuroscience** (study of the brain and nervous system), **placebo effect** (improvement caused by expectation), **single-blind study** (only participants are unaware of their group).

PART 2 — THE BOOK’S OWN QUESTIONS, WITH OFFICIAL ANSWERS

Show What You Know — “What Is Psychology and How Did It Begin?”

1. The goal of ______ is to gather knowledge for the sake of knowledge, whereas the goal of ______ is to change behaviors and outcomes. → **basic research; applied research**
2. A college dean contacts the psychology department to design a program to encourage students to study together, hoping to increase retention. Which main goal of psychology? → **d. change**

3. William James suggested it is important to study the purpose of thoughts, feelings, and behaviors and how they help us adapt — this theme is: → **b. functionalism**
4. Pick two perspectives and describe how they are similar; pick two and explain how they differ. → *Sample official answer:* **Sociocultural and biopsychosocial are similar** in that both examine how interactions with other people influence behaviors and mental processes. **Cognitive and biological differ** in that cognitive focuses on the thought processes underlying behavior, while biological emphasizes physiological processes.

Show What You Know — "How Do Psychologists Do Research?"

1. A research group randomly selects students from across the nation, trying to pick a ______ that closely reflects the characteristics of American college students. → **c. representative sample**
2. Researchers must establish ______ that specify the precise manner in which characteristics of interest are defined and measured. → **operational definitions**
3. Why might it be problematic to make inferences from small samples to large populations? → *Official answer:* With a large population, a researcher needs a sample; a random sample gives every member an equal chance of selection and increases the likelihood of a representative sample (a subgroup whose characteristics are similar to the population's). If the sample is small relative to the population, we must be careful drawing inferences from sample to population; representative samples help ensure inferences are meaningful.

Show What You Know — "Descriptive and Correlational Methods"

1. Descriptive research is primarily useful for studying: → **c. new or unexplored topics**
2. A researcher finds a positive correlation between smartphone use and absenteeism from work — why can't she say phone use *causes* absenteeism? → *Official answer:* The correlational method illuminates relationships and helps make predictions, but cannot determine cause and effect. Even strongly correlated variables may be driven by a third variable — possible third variables here include boredom on the job, an ill family member, or apathy about work.
3. A researcher trains teachers to use a stopwatch to record how long caregivers take to enter and leave the classroom at drop-off — this approach to collecting data is: → **naturalistic observation**

Show What You Know — "The Experimental Method"

1. Following a double-blind study with a treatment and placebo, the researcher meets each participant to discuss important information about the study. This is known as: → **b. debriefing**

2. Describe what a double-blind study is and why deception is necessary. → *Official answer:* In a double-blind study, neither participants nor the researchers working directly with them know who is getting the treatment and who is getting the placebo. It reduces expectations and biases that can arise when either side knows what they've received or distributed — so that neither experimenter nor participant consciously or unconsciously alters behavior and changes the outcome.

Test Prep — Are You Ready? (end-of-chapter, with official answers)

1. Experts used prior research findings to create a campaign to curb binge drinking among students. This is an example of: → **b. applied research**
2. Students weigh evidence from several journal articles, synthesize the information, and determine each finding's contribution to understanding a topic. This process is: → **b. critical thinking**
3. Plato believed truth and knowledge exist in the soul before birth. This supports: → **c. the nature side of the nature–nurture issue**
4. ______ suggests that human nature is by and large positive. → **d. Humanistic psychology**
5. The goal of ______ is to provide empirical evidence based on systematic observation or experimentation. → **c. the scientific method**
6. A psychologist studying identical twins was interested in their leadership qualities and educational backgrounds. These characteristics are generally referred to as: → **c. variables**
7. One way to pick a random sample is to make sure every member of the population has: → **d. an equal chance of being picked to participate**
8. A researcher interested in the effect of separating identical twins shortly after birth might use a(n) ______, a type of descriptive research invaluable for studying rare events. → **b. case study**
9. With a(n) ______ study, neither the researchers nor the participants know who is getting the treatment or the placebo. → **a. double-blind**
10. A researcher forming a ______ of high school seniors might select participants using a comprehensive database of all seniors, so that all members of the population have an equal chance of being selected. → **d. random sample**
11. Describe the goals of psychology and give an example of each. → *Official answer sketch:* Describe, explain, predict, and change behavior — e.g., a researcher might conduct a study to describe differences in children's food preferences; seeing patterns within families, she might explain them via the home environment; she might predict food preferences from close family members; and she might apply the findings to help experts create a campaign for healthy eating.

12. A researcher planning a study on aggression and exposure to media violence — how can she ensure ethical treatment of the children in her study? → *Official answer sketch:* Read how other studies on this topic ensured ethical treatment; study the ethical guidelines of the professional organizations; and submit her proposal to the Institutional Review Board.
13. Use the perspectives of psychology to explain the similarities between Sam and Anaïs. → **See Table 1.1** — e.g., biological/evolutionary emphasize shared genes; behavioral and sociocultural emphasize learned similarities and cultural contexts; biopsychosocial integrates all of these.
14. Find an article in the popular media that presents variables as having a cause-and-effect relationship when it's really correlational. → *Hint from the chapter:* look for topics that would be very hard to manipulate ethically — breast feeding, amount of television watched, attitudes.
15. Reread the SpongeBob study. How does it establish cause and effect? What would you change or improve on replication? → *Official answer sketch:* The researchers used the experimental method: children were randomly assigned to groups, researchers manipulated the activities (IV), other variables were held constant, so differences in cognitive performance (DV) can be attributed to the manipulation. One way to change the study: use a different age group.

PART 3 — SPOKEN-REVIEW QUIZ PROMPTS (for the podcast to use)

History and people: 1. What year, where, and by whom was the first psychology laboratory founded? *(1879, University of Leipzig in Germany, Wilhelm Wundt.)* 2. Who is called the father of psychology and what was his method? *(Wundt; introspection — objective examination of one's own conscious activities.)* 3. Structuralism vs. functionalism — who founded each, and what did each ask? *(Titchener: what is the mind made of; James: what are thoughts, feelings, behaviors FOR — adaptation.)* 4. Which woman earned the first psychology PhD and where? Which was denied hers and where — and what "first" did she achieve anyway? *(Washburn — Cornell, 1894; Calkins — Harvard denied it, first female APA president, 1905.)* 5. Whose research helped decide Brown v. Board of Education? *(Mamie Phipps Clark, with Kenneth Clark.)* 6. Match: classical conditioning / operant conditioning / founding behaviorism. *(Pavlov / Skinner / Watson.)* 7. Which two people founded humanistic psychology and what was it reacting against? *(Rogers and Maslow; the determinism of psychoanalysis and behaviorism.)* 8. Nature or nurture: Plato? Aristotle? Locke? The behaviorists? *(Nature; nurture; nurture; nurture.)* 9. What is the "Freud Problem"? *(Freud's fame vastly exceeds the scientific support for his* *theories; ~90% of APA members don't practice psychoanalysis.)* 10. Which perspective goes with: Vygotsky? David Buss? George Miller? *(Sociocultural; evolutionary; cognitive.)*

Methods: 11. Name the five steps of the scientific method in order. *(Question → hypothesis → design & collect → analyze → publish.)* 12. Random sample vs. random assignment — when does each happen? *(Sampling recruits participants from the population at the start; assignment divides recruited participants into groups later.)* 13. Why can't correlation prove causation? Give both reasons. *(Third variables; directionality.)* 14. r = –.85 vs. r = +.20 — which is stronger, and what direction is each? *(–.85 is stronger; negative/inverse. +.20 is weak positive.)* 15. In the caffeine study: name the IV, DV, one possible extraneous variable, and one possible confound. *(Caffeine vs. sugar pill; attention-task performance; caffeine sensitivity ruining sleep; time of day of testing.)* 16. What's the difference between a single-blind and double-blind study, and what bias does each control? *(Single: participants unaware — controls participant expectations/placebo effects; double: participants AND administering researchers unaware — also controls experimenter bias.)* 17. What did Clever Hans actually demonstrate? *(Experimenter bias — the horse read unintentional human cues; expectations leak into results.)* 18. SpongeBob study: sample, three conditions, DV, result, and one limitation. *(Sixty 4-year-olds, mostly White upper-middle-class; SpongeBob vs. educational show vs. drawing, 9 minutes each; executive-function tests; SpongeBob group performed worse; limits — temporary effect, unrepresentative sample.)* 19. Which research method would you choose to study: a rare twin reunion? attitudes of a million students? whether a new drug causes improvement? *(Case study; survey; experiment.)* 20. A case study can never support or refute a hypothesis — why? *(It's a sample of one; hypothesis testing requires comparisons between conditions.)*

Ethics and beyond: 21. Name the five principles the APA/APS/BPS guidelines encourage. *(Do no harm; safeguard welfare of humans and animals; know responsibilities to society/community; maintain accuracy; respect human dignity.)* 22. Informed consent vs. debriefing — before or after, and what does each cover? *(Consent before: procedures, risks, benefits, confidentiality, right to withdraw; debriefing after: purpose and any deception.)* 23. When MUST a therapist break confidentiality? *(Danger to self or others; abuse, neglect, or domestic violence; court order.)* 24. What board must approve all human and animal research? *(The Institutional Review Board — IRB.)* 25. What is positive psychology and which earlier perspective set its stage? *(The study of human strengths and the roots of happiness, creativity, love; humanism.)* 26. What happened with the vaccine–autism study, and which two safeguards of science did the episode showcase? *(Wakefield's 1998 study was fraudulent and retracted after 12 years; replication failures exposed it and peer review/retraction corrected* *the record — though public damage lingers.)* 27. Why do identical twins matter so much to psychology? *(Same genes — so similarities despite different environments reveal nature's contribution, and differences reveal nurture's; twin studies untangle nature and nurture for intelligence, personality, health.)*